PSI Data Organization Document

Utilization of the Smoke Aerosol Measurement Experiment Data for Advanced Modeling and Simulation of Smoke Generation in Micro-Gravity

To gain an initial understanding of this experiment, refer to the Final Report, located on "Pub and Docs" tab of this Investigation Record. The simulation data is organized in PSI in "Experiment Data" on the "Data" tab, in two high level categories:

- Earth Gravity Figure Data
- Microgravity Figure Data

The simulation data is further categorized into specific cases:

- Case 1 with inlet velocity: U_o = 0:08 m/s
- Case 2 with inlet velocity: U_o = 0:1 m/s

Additional test parameters and conditions are located in Table 3 of the Final Report:

Table 3: Definitions of primary test cases.

Case	$U_o[\mathrm{m/s}]$	$g_o[\mathrm{m/s^2}]$	Re	Fr
1EG	0.08	9.81	22.3	0.385
1MG	0.08	0.0	22.3	∞
2EG	0.10	9.81	27.7	0.481
2MG	0.10	0.0	27.7	∞

Where EG = Earth Gravity; MG = Microgravity; U_o = Inlet Velocity; g_o = Gravitational Acceleration; *Re* = Reynolds number; *Fr* = Froude number.

In addition to the four base cases listed in Table 3 of the Final Report, and as shown above, the data is further categorized as the researcher examines the aerosol development sensitivity to:

- Particle settling
- Thermophoresis
- The flow geometry

For this purpose, a series of additional cases, defined in Table 3 of the Final Report, and as shown above, are constructed by modifying the setup of Case 1MG and/or Case 1EG.

- **Case1-Earth Gravity-NS**: (where NS stands for "no settling") is identical to Case1-Earth Gravity except the settling mechanism has been neglected.
- **Case1-Earth Gravity-R**: (where R stands for "rotated") is identical to Case1-Earth Gravity except the heated vapor source has been rotated 90°.
- **Case1-Microgravity-NT**: (where NT stands for "no thermophoresis") is identical to Case1-Microgrativy except the thermophoresis transport mechanism has been neglected.
- **Cases1Microgravity-R**: (where R stands for "rotated") is identical to Case1-Microgrativy except the heated vapor source has been rotated 90°.

In all cases, data is collected and presented at a time of t = 7 seconds.